

Ammer Saeed

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PROFESSIONAL OVERVIEW

AI Engineer specialising in on-device LLM deployment, agentic AI systems, and AI-powered cross-platform products. Proficient in integrating local and cloud LLMs (Mistral 7B, Gemini AI) with Flutter, Rust, and React across production-grade systems. Driven by the intersection of AI, real-time infrastructure, and mental health technology.

EDUCATION

Bachelor of Science in Software Engineering

2023 – 2027

National University of Sciences and Technology (NUST), Islamabad, Pakistan

PROJECTS

Sehat – AI Clinical Intake Triage Agent

sehat-six.vercel.app | github.com/OmgItsAmmer/Sehat

- Engineered a **multi-node LangGraph state machine** (classify, probe, slot-fill, route, book, escalate) with **confidence-thresholded human-in-the-loop interrupts** and hardcoded P1 keyword overrides for cardiac and respiratory emergencies, as measured by **sub-4-second P1 alert latency** and **80%+ autonomous case resolution** across bilingual Urdu/English WhatsApp and web channels.
- Designed a **RAG pipeline** using **pgvector** + **text-embedding-3-small** over a structured clinic knowledge base with **specialist sub-agent routing** (cardiology, pediatrics, general) and **real-time appointment slot management** via natural conversation — eliminating manual receptionist intake for **P2/P3 cases end-to-end** from symptom capture to confirmed booking with automated reminders.
- Delivered a **production-grade FastAPI backend** with **PostgreSQL** + **Redis session persistence**, **Alembic migrations**, and a **React receptionist dashboard** with real-time override audit logging, validated by **119 automated tests** (unit, integration, system) triggered via **CI/CD pipelines** and a **20-message bilingual classification eval suite** reporting per-class pAUC across all urgency levels.

okiosk – Agentic AI Kiosk Platform

okiosk.vercel.app | github.com/OmgItsAmmer/okiosk

- Architected a **full-stack agentic AI kiosk platform** using Flutter (mobile/desktop), React web kiosk, and a **Rust/Axum backend**, enabling natural-language cart and search actions via a local **Mistral 7B LLM** and cloud LLM integration, reducing manual kiosk interaction time significantly.
- Delivered **real-time kiosk synchronization** via **Socket.IO WebSockets**, implemented **Whisper.cpp** speech-to-text for voice-driven assistant flows, and secured sessions with **Google OAuth** + **JWT-based guest tokens**, supporting concurrent multi-user kiosk sessions.
- Engineered a **race-condition-safe checkout API** in **Supabase/PostgreSQL** with stock validation and **QR-based kiosk login**, eliminating inventory inconsistency errors across simultaneous transactions.

NUST Admissions Assistant – Hybrid Offline-First RAG Chatbot

github.com/OmgItsAmmer/nust_faq_bot

- Engineered a **hybrid RAG pipeline** combining **Ollama Llama 3.2 3B** (on-device inference) with **Gemini Cloud API** as fallback, using **Ollama nomic-embed-text** for local embeddings and **ChromaDB** for vector search, achieving **offline-first** operation with sub-5-second end-to-end response times on 71 indexed FAQ entries.
- Delivered **multilingual admissions support** (English, Urdu, and code-switched queries) across **multiple channels** via **OpenClaw** plugin architecture, routing natural-language questions through a **FastAPI** service that maps Gemini quota and runtime errors to structured HTTP responses, ensuring graceful degradation at every layer.
- Reduced query-time overhead to **3 inference steps** (embed, retrieve, generate) by pre-indexing all documents offline with **reindex.py**, decoupling embedding model updates from runtime and enabling zero-downtime re-indexing on model or dimension changes.

WORK EXPERIENCE

Rust Developer – BlueDiamondAI, Germany (Remote)

July – September 2024

- Contributed to **LLM operations tooling in Rust**, improving inference pipeline efficiency, as measured by reduced overhead in model-serving scripts, by applying Rust's zero-cost abstractions to ML infrastructure tasks.

ML Engineer Intern – Muraabi Lab, SINES NUST, Pakistan

June – September 2024

- Developed an **interactive analytics dashboard** using **Power BI and Tableau**, as measured by enabling non-technical stakeholders to self-serve insights, by transforming raw datasets into filterable, visual reporting layers.
- Built a **skin cancer detection model** using **deep learning (CNN)** achieving clinically meaningful classification accuracy, as measured by model performance metrics on a curated medical image dataset, by fine-tuning a pre-trained vision model.

TECHNICAL SKILLS

Languages: Rust, Dart, JavaScript, Python, SQL

Frameworks: Flutter (Mobile/Web/Desktop), React, Express, Axum (Rust)

AI/ML: Mistral 7B (on-device), Gemini AI, Whisper.cpp, CNN (deep learning), LLMOps

Databases & Backend: Supabase, PostgreSQL, Firebase, Supabase Edge Functions, Socket.IO

Tools: Docker, Google Cloud, Power BI, Tableau, GitHub

LEADERSHIP

President – The Hustle Hour (SEECs-Level Speaker Event Series)

Led end-to-end execution of a high-profile speaker series featuring founders from **Vector**, **Inlights**, **Markaz**, and **SearchOPal**, demonstrating cross-functional leadership by managing outreach, logistics, on-ground coordination, and securing sponsorships from **Vital**, **Asia Ghee**, **Next Cola**, **Times**, and **Capital Pizza** – delivering a polished, multi-stakeholder event under real organizational constraints.